

Toward Rational Design of Selective Molecularly Imprinted Polymers (MIPs) for Proteins: Computational and Experimental Studies of Acrylamide Based Polymers for Myoglobin

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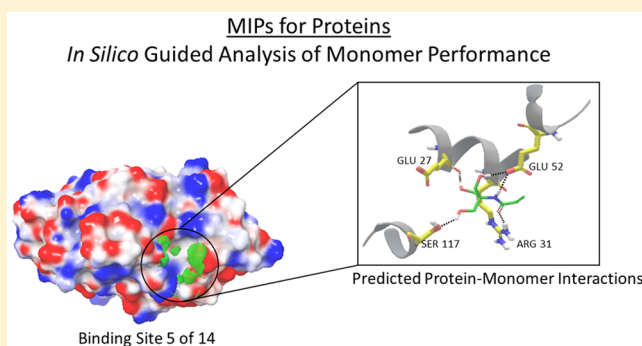
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Supporting Information

ABSTRACT: Molecularly imprinted polymers (MIPs) have potential as alternatives to antibodies in the diagnosis and treatment of disease. However, atomistic level knowledge of the prepolymerization process is limited that would facilitate rational design of more efficient MIPs. Accordingly, we have investigated using computation and experiment the protein–monomer binding interactions that may influence the desired specificity. Myoglobin was used as the target protein and five different acrylamide-based monomers were considered. Protein binding sites were predicted using SiteMap and binding free energies of monomers at each site were calculated using MM-GBSA. Statistical thermodynamic analysis and study of atomistic interactions facilitated rationalization of monomer performance in MIP rebinding studies (% rebind; imprinting factors). CD spectroscopy was used to determine monomer effects on myoglobin secondary structure, with all monomers except the smallest monomer (acrylamide) causing significant changes. A complex interplay between different protein–monomer binding effects and MIP efficacy was observed. Validation of hypotheses for key binding features was achieved by rational selection of two different comonomer MIP combinations that produced experimental results in agreement with predictions. The comonomer studies revealed that uniform, noncompetitive binding of monomers around a target protein is favorable. This study represents a step toward future rational *in silico* design of MIPs for proteins.



1. INTRODUCTION

Molecular imprinting is fast becoming an established technique to produce polymers with molecular recognition properties.^{1–4} These molecularly imprinted polymers (MIPs) are inspired by nature and are capable of binding various analytes with an affinity and selectivity similar to those of their biological equivalents. The advantages of MIPs over biological receptors includes durability, specificity, and ease of mass production.^{5–7} They have broad applications in areas such as chromatography, solid phase extractions, catalysis, and (bio)sensors.^{8–11} While protein MIPs have considerable potential to replace antibodies in the diagnosis and treatment of disease, they still have limitations. MIPs currently lack the selectivity required for practical applications partly due to a poor understanding of how to optimize protein–MIP interactions at cavity sites. The MIP cavities are borne out of the noncovalent interactions within the prepolymerization mixture, leading to binding site heterogeneity especially within bulk MIP.⁹ This means that a better understanding of the protein–monomer interactions that are

favorable for producing selective MIPs is needed to progress them further as a viable alternative to their biological counterparts. There has been a particular increase in macromolecular imprinting over the past decade, mainly due to demand in the areas of biosensors, clinical diagnostics and therapeutics.^{12,13} Furthermore, toward an alternative to the expensive and degradable antibodies currently being used in analysis,^{14–17} there is particular interest in selectively imprinting macromolecules such as proteins, viruses, and cells, for their subsequent determination within a complex medium. Employing hydrogels for the imprinting of proteins has shown considerable promise toward this aim.¹⁸ The hydrogels consist of significantly dilute cross-linked systems that form an insoluble network of hydrophilic cross-linked polymer chains.

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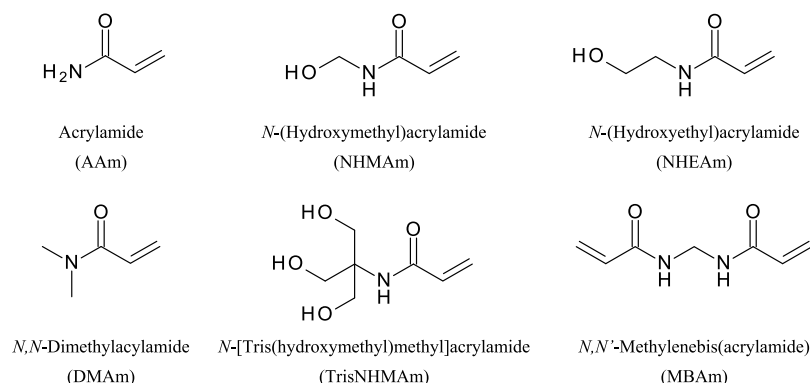


Figure 1. Five monomers (AAM, NHMAm, NHEAm, DMAm, TrisNHMAm) and cross-linker (MBAm) used in the polymerization of the hydrogel MIPs.

MIP formation is achieved when a mixture consisting of functional monomers and cross-linker polymerizes around a target molecule.^{7,19,20} The target molecule in effect acts as a template for MIP formation. After polymerization, the template molecule is removed, to leave binding sites on the newly formed MIP capable of selectively recognizing the target molecule.^{21,22} Molecular imprinting is a consequence of monomer–template interactions, with noncovalent interactions such as hydrogen bonding, van der Waals, and ionic interactions, together with hydrophobic effects used in the self-assembly process of the prepolymerization complex.^{23–25} In order to achieve a successful imprinting procedure, there needs to be a strong monomer–template complex formed during prepolymerization, which is preserved through to the final polymerization step, meaning the choice of monomer is of paramount importance.^{4,26} A more stable monomer–template complex with strong intermolecular interactions can potentially deliver a MIP with high selectivity and affinity.²² In this regard, *in silico* predictive methods can provide a rationale for improved monomer–template complexation.

To date, however, computational methods have been mainly applied to the molecular imprinting of low molecular weight target (analyte) molecules. Molecular modeling software and searching algorithms, traditionally applied in drug design, have been adapted to find monomers with the ability to form strong complexes with small molecule templates.²⁷ Virtual libraries of polymerizable, functional monomers have been created and used to successfully predict *in silico* selective MIPs for small molecule analytes such as ephedrine.^{27,28} Computational methods allow monomers to be ranked in terms of predicted binding affinity. For example, the highest ranked monomers from LUDI docking²⁹ were used to produce a rationally designed selective MIP for microcystin-LR, a toxic analyte in water.³⁰ Molecular dynamics (MD) have applications in modeling multicomponent systems and assessing a large number of interactions²⁷ and have been used to simulate and evaluate MIP prepolymerization mixtures.³¹ These studies allowed a closer analysis of how the functional monomers, cross-linker, and solvent interacted with the template and the preferred arrangement of interactions in the prepolymerization mixtures.³¹ Quantum mechanics (QM) methods are also becoming more common place in MIP design, although limited by the size of the studied systems and associated computational expense.²⁷ QM in the form of density functional theory (DFT) calculations was successfully applied for the selection of monomers to design MIPs for 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD), a well-known toxin.³²

There have been few computational studies using proteins as templates for MIPs.³³ Due to the size of proteins, a high molar ratio of ligand to protein is required, with ratios of 2000:1 commonly seen. Molecular docking can predict where the ligands prefer to bind.³⁴ Docking studies of common monomers and cross-linkers to target proteins have highlighted complementary shape and demonstrated that there is a considerable amount of hydrogen bonding, electrostatic and hydrophobic interactions between the target and monomers for specific recognition.^{34,35} However, in addition to favorable interactions, there are also monomer interactions with the target protein backbone that have the potential to cause secondary structural changes affecting MIP selectivity.³⁴ Boroznjak et al. used a docking approach to rationally select a functional monomer from *m*-phenylenediamine (mPD), dopamine, and 3,4-ethylenedioxythiophene for building a MIP suitable for immunoglobulin G (IgG).³³ The three monomers were docked into the fragment antigen-binding (FAB), fragment crystallizable (Fc), and hinge regions of IgG. All three monomers studied revealed similar docking scores, but mPD molecules were found to be more uniformly arranged around IgG, leading to a MIP with more complementary cavities, and thus, better experimental performance.

In this work, supported by experimental data, we have investigated at the atomic level using computation the features that govern the binding and relative efficacies of different monomers for use in MIPs demonstrating selective target protein binding. The model target protein for these studies was myoglobin, while five acrylamide-based monomers (Figure 1) were chosen that would allow the analysis of different structural effects. Myoglobin is a protein released during damage to muscle tissue. Clinically, it can be a sensitive marker for muscle injury and has potential to be used as a cardiac marker for myocardial infarction. Importantly, for the purposes of this study, because of the protein's small size (153 residues, ~17 kDa) and with fewer monomer binding sites, interpretation and analysis of the modeling results is better facilitated. It should be noted that the more readily available equine isoform of the protein was used for analysis, which has high homology with human form (88% identity),³⁶ so that results in terms of actual medical applications would need to be confirmed. The potential binding sites of the protein were predicted using the SiteMap program,^{37–39} followed by prediction of the binding site-specific interactions of each monomer studied using Glide docking and postdocking molecular mechanics with generalized Born and surface area continuum solvation (MM-GBSA) binding free energy (BFE) calculations.³⁹

Docking has previously been applied for studies of this type but mainly for small molecule analytes (as highlighted above), while postdocking MM-GBSA should theoretically improve the accuracy of predictions. A comprehensive statistical thermodynamic analysis and study of protein–monomer interactions allowed us to apply a rational approach to predict comonomer combinations that produce hydrogel MIPs with different efficacies compared to those MIPs of the individual monomers. Additionally, the effects of monomers on protein secondary structure features were analyzed by far-UV circular dichroism (CD) spectroscopy, with the SELCON3 method, as this can provide an accurate estimation of protein secondary structure.^{40,41}

2. EXPERIMENTAL SECTION

2.1. Computational details. **2.1.1. Protein and Monomer Preparation.** The initial setup of the myoglobin protein (PDB code: 5DSR) containing an iron-centered heme group for calculations was performed using Schrödinger's "Protein Preparation Wizard".³⁹ Water molecules were deleted, bond orders were assigned, and hydrogen atoms added, with protonation states for basic and acidic residues based on residue pK_a values at normal pH (7.0) calculated using PROPKA.⁴² Subsequent optimization of hydroxyl groups, histidine protonation states and C/N atom flips, and side-chain O/N atom flips of Asn and Gln was based on optimizing hydrogen bonding patterns. Finally, an "Impref" minimization of the myoglobin was performed using the OPLS3 force field⁴³ to remove steric clashes and bad contacts, but with heavy atoms constrained to within 0.3 Å (RMSD) of their crystallographic positions. The monomers shown in Figure 1 were prepared for calculations using LigPrep 3.5³⁹ with the optimized potential for liquid simulations 3 (OPLS3) force field⁴³ and default settings.

2.1.2. Protein Binding Site Analysis. Potential monomer binding sites on myoglobin were predicted using SiteMap 3.9.^{37,39} The settings included at least five site points per site, a "less restrictive" definition of hydrophobicity, along with a standard grid. Shallow binding sites were also detected. The sites generated were cropped at a distance of 3 Å from the nearest site point, with a total of 14 predicted binding sites identified. The surface of the protein was analyzed by using BioLuminate 16.2 with protein surface analyzer.³⁹

2.1.3. Docking Calculations. For each predicted myoglobin binding site, docking calculations on the five monomers were performed using Glide 7.1 in standard-precision (SP) mode^{39,44,45} with default OPLS3 atomic charges and van der Waals scaling (0.8) for ligand nonpolar atoms to include modest "induced fit" effects. Docking grids were generated for each of the 14 different potential binding sites from SiteMap, with the grid center coordinates included in Table S1. Postdocking minimization of poses with strain correction was employed. Up to 20 poses per input monomer structure were saved for each docking run, in order to generate a large number of diverse poses.

2.1.4. Post-Docking MM-GBSA Calculations. Using the Glide-SP docking poses for each monomer at each site, MM-GBSA binding free energies were calculated using Prime 4.4³⁹ and the following equation:

$$\Delta G_{\text{PL}} = \Delta E_{\text{PL}}^{\text{MM}} + \Delta G_{\text{PL}}^{\text{sol}} - T\Delta S_{\text{PL}}^{\text{MM}} \quad (1)$$

where $\Delta E_{\text{PL}}^{\text{MM}}$ represents the molecular mechanics (MM) energy difference (internal, electrostatic and van der Waals)

between the protein (P)–monomer (L) bound and unbound states, calculated using the OPLS3 force field.⁴³ $\Delta G_{\text{PL}}^{\text{sol}}$ is the corresponding solvation free energy change on binding calculated using a variable-dielectric generalized Born solvation model. While the protein was fixed for these calculations, ligand strain (reorganization) effects on binding were included. Estimates for solute entropy changes on binding $\Delta S_{\text{PL}}^{\text{MM}}$ were also included in terms of the change in vibrational, rotational and translational (VRT) entropy of ligands on binding. The latter was calculated using the rigid rotor harmonic oscillator (RRHO) approximation (default settings) with MacroModel v11.3³⁹ and the OPLS3 force field.⁴³

2.1.5. Statistical Thermodynamics Analysis of Monomer Binding. In order to analyze the probabilities for monomers to bind at each of the predicted binding sites in the prepolymerization state, we performed statistical thermodynamic analysis of the ΔG_{PL} data. A full derivation of the employed formulas is detailed in the Supporting Information. We consider a protein P with M independent, nonequivalent binding sites, indexed by the integers $i = \{1, \dots, M\}$. We assume that each protein site can be empty or bind one monomer. Binding of monomer L at site i is associated with a solution free energy change $\Delta G_{\text{PL}}(i)$.

Single Monomer Studies. We first consider the case of a solution with dissolved protein and only one monomer species L. The concentration of free monomer in solution, $x \equiv [L]/C_0$, measured in units of a standard-state concentration C_0 , is computed from the self-consistent solution of the following equation (eq 3 of the Supporting Information):

$$\frac{[L]}{C_0} = x = \frac{[L]_{\text{tot}}}{C_0} - \frac{[P]_{\text{tot}}}{C_0} \sum_{i=1}^M \frac{x e^{-\beta \Delta G_{\text{PL}}(i)}}{1 + x e^{-\beta \Delta G_{\text{PL}}(i)}} \quad (2)$$

where β equates to $1/kT$. $[L]_{\text{tot}}$ is the total ligand concentration (0.76 M) and $[P]_{\text{tot}} = [L]_{\text{tot}}/1081 = 7.03 \times 10^{-4}$ M is the total protein concentration. A similar equation for equivalent sites is derived in ref 46.

The average number of monomer L molecules bound to a protein molecule as a function of the free ligand concentration x is [SI eq 2]:

$$\bar{M}_L(x) = \sum_{i=1}^M \frac{x e^{-\beta \Delta G_{\text{PL}}(i)}}{1 + x e^{-\beta \Delta G_{\text{PL}}(i)}} \quad (3)$$

Comonomer Studies. We now consider comonomer solutions, in which two monomers L_1 and L_2 are simultaneously present in the solution. The free-monomer concentrations are evaluated by solving self-consistently the following system of equations [Supporting Information eq 6]:

$$\begin{aligned} x_1 &= \frac{[L_1]_{\text{tot}}}{C_0} - \frac{[P]_{\text{tot}}}{C_0} \sum_{i=1}^M \frac{x_1 e^{-\beta \Delta G_{\text{PL}_1}(i)}}{1 + x_1 e^{-\beta \Delta G_{\text{PL}_1}(i)} + x_2 e^{-\beta \Delta G_{\text{PL}_2}(i)}} \\ x_2 &= \frac{[L_2]_{\text{tot}}}{C_0} - \frac{[P]_{\text{tot}}}{C_0} \sum_{i=1}^M \frac{x_2 e^{-\beta \Delta G_{\text{PL}_2}(i)}}{1 + x_1 e^{-\beta \Delta G_{\text{PL}_1}(i)} + x_2 e^{-\beta \Delta G_{\text{PL}_2}(i)}} \end{aligned} \quad (4)$$

The average numbers of monomers, L_1 or L_2 , bound to a protein molecule are computed by the relations [Supporting Information eqs 5]:

$$\bar{M}_{L_1}(x_1) = \sum_{i=1}^M \frac{x_1 e^{-\beta \Delta G_{\text{PL}_1}(i)}}{1 + x_1 e^{-\beta \Delta G_{\text{PL}_1}(i)} + x_2 e^{-\beta \Delta G_{\text{PL}_2}(i)}} \quad (5a)$$

and

$$\bar{M}_{L_2}(x_2) = \sum_{i=1}^M \frac{x_2 e^{-\beta \Delta G_{PL_2}(i)}}{1 + x_1 e^{-\beta \Delta G_{PL_1}(i)} + x_2 e^{-\beta \Delta G_{PL_2}(i)}} \quad (5b)$$

Finally, the relative probabilities of monomers L_1 and L_2 to bind at the same protein site k (in the cosolvent solution) are calculated as [Supporting Information eq 14]:

$$\frac{p(\text{site } i \text{ occupied by } L_1)}{p(\text{site } i \text{ occupied by } L_2)} = \frac{x_1 e^{-\beta \Delta G_{PL_1}(i)}}{x_2 e^{-\beta \Delta G_{PL_2}(i)}} \cong \frac{e^{-\beta \Delta G_{PL_1}(i)}}{e^{-\beta \Delta G_{PL_2}(i)}} \quad (6)$$

The last equality is valid when the monomers have the same free concentration ($x_1 \cong x_2$).

2.2. MIP Synthesis. **2.2.1. Materials.** Acrylamide (AAm), ammonium persulfate (APS), α -lactalbumin, glacial acetic acid (AcOH), myoglobin (from equine skeletal muscle) (Mb), *N*-(hydroxymethyl)acrylamide (NHMAm), *N*-(hydroxyethyl)acrylamide (NHEAm), *N,N*-dimethylacrylamide (DMAm), *N,N'*-methylenebis(acrylamide) (mBA), *N*-[tris(hydroxymethyl)methyl]acrylamide (TrisNHMAm), sodium dodecyl sulfate (SDS), and tetramethylethyldiamide (TEMED), were all purchased and used without purification from Sigma-Aldrich, Poole, Dorset, UK.

2.2.2. Methods. Solution Preparation. A solution of 10% (w/v):10% (v/v) SDS:AcOH was prepared for use in the washing (protein elution) stages before the template reloading stage. SDS (10 g) and AcOH (10 mL) were dissolved in 990 mL of deionized (DI) water, to produce 1 L of the washing solution.

MIP Preparation. Bulk MIP hydrogels were produced, using an optimized methodology,¹⁸ where a 10% cross-linking monomer/*N,N'*-methylenebis(acrylamide) hydrogel was found to produce the optimal imprint for Mb, in terms of specificity and rebinding efficiency of the MIP, compared with the nonimprinted polymer (NIP).¹⁸

The MIPs were produced with different monomers (AAm, NHMAm, NHEAm, DMAm, TrisNHMAm) and a 10% cross-linking density for the myoglobin as a template using the following method. Into an Eppendorf tube, 12 mg of myoglobin template was dissolved in 970 μ L of deionized water vortexed for 1 min, followed by the addition of 7.6×10^{-4} mol of functional monomer and mBA (cross-linker) at a ratio of 9:1 by weight, then vortexed for a further minute. Finally, 10 μ L of a 5% TEMED (v/v) solution and 20 μ L 5% APS (w/v) solution were added and the mixture was vortexed for 1 min. Solutions were purged with nitrogen for 5 min and polymerization occurred overnight at room temperature ($\sim 20^\circ\text{C}$). Corresponding NIPs were produced using the same method, but in the absence of a protein template.

After polymerization, the gels were granulated separately using a 75 μ m sieve. The refined gels were washed with five 1 mL volumes of deionized water followed by five 1 mL volumes of 10% (w/v):10% (v/v) SDS:AcOH eluent; this allowed for the removal of the template protein from the MIP cavities. Following this, the gels were washed with five 1 mL volumes of deionized water to remove all residual 10% (w/v):10% (v/v) SDS:AcOH from the MIP gels. Each wash step was followed by centrifugation, whereby the gels were vortexed and then centrifuged (using SLS Lab basics centrifuge) for 5 min at 15000 rpm (RCF: 15100g). Corresponding nonimprinted polymers (NIPs) were synthesized

using the same procedure as the MIPs, but in the absence of the template molecule.

2.2.3. MIP Rebinding Studies. The subsequent rebinding effect of the conditioned and equilibrated MIPs and NIPs were characterized using the BioDrop μ LITE UV/visible spectrometer. Hydrogels (200 mg) were then each treated with 400 μ L of a 3 mg mL⁻¹ myoglobin (template protein) solution. The polymer/protein solutions were mixed on a rotary vortex mixer and allowed to associate at room temperature ($\sim 20^\circ\text{C}$) washed four times with 1 mL of deionized water. Each reload and wash step for hydrogels was followed by centrifugation for 5 min at 15000 rpm (RCF: 15100g). All supernatants were collected for analysis by spectrophotometry (at 405 nm for myoglobin).

2.3. CD Spectroscopy. The secondary structure of myoglobin was determined using a J-815 JASCO circular dichroism spectropolarimeter at 20°C . Essentially, protein samples (0.00075 mg/mL) were spiked with monomer solution to produce a molar ratio of 1:1081 and then placed in a quartz cell (Starna Scientific UK) with a 10 mm path length. Using a CD spectrometer, far-UV CD spectra were collected for the protein samples, where the instrument was set to standard, ten scans per sample were performed over a wavelength range of 260 to 180 nm at a bandwidth interval of 1 nm, data pitch 0.5 nm, scanning mode set to continuous and a scan speed set at 100 nm min⁻¹. All spectra were baseline corrected and the percentage secondary structure was determined using the Selcon3 method (protein reference set 3) from the DichroWeb server.^{47–49} These experiments were repeated three times and the secondary structure percentages were averaged.

3. RESULTS AND DISCUSSION

3.1. Single Monomer Computational and Experimental Studies. **3.1.1. Single Monomer MIPS Experimental Results.** Each of the five monomers in Figure 1 (AAm, NHMAm, NHEAm, DMAm, TrisNHMAm) was used to prepare five different MIPs and their corresponding NIPs, where the NIPs were control polymers for MIP selectivity. The NIPs were produced in an identical manner to the MIPs, but in the absence of the protein template, and thus contained the same chemical properties as a MIP, but without the specific cavities. Rebinding studies were performed on the MIPs and NIPs, and the percentage of target protein rebind and subsequent imprinting factors (IFs) were calculated using the following equation:

$$\text{IF} = \frac{\% \text{ protein rebind to MIP}}{\% \text{ protein bind to NIP}} \quad (7)$$

Thus, IF is calculated as a ratio of protein rebind to the MIP versus protein that binds to the corresponding NIP. IF is commonly used to evaluate the imprinting effect and to measure the strength of interaction between the functional monomer and the template/target molecule. The higher the IF, the more selective the MIP is for the target molecule, and an IF > 1.20 is generally considered favorable.⁹ Whereas it is accepted that a selectivity factor (ratio of MIP binding to target versus a nontarget protein) is a better determinant of selectivity, use of this factor is beyond the scope of this present study.

The single monomer experimental results (Table 1) revealed that NHMAm was the best monomer used to produce a MIP

Table 1. Percentage of the Myoglobin Target Protein Rebind to the Five Different Acrylamide Based MIPs and NIPs, and Their Corresponding Impact Factors^a

monomer/ comonomer	MIP percentage of protein rebind (%)	NIP percentage of protein bind (%)	imprinting factor (IF)
NHMAm	98.9 ± 0.2	51.8 ± 0.4	1.90
AAM	85.4 ± 1.0	47.5 ± 4.2	1.80
NHEAm	77.2 ± 3.0	43.6 ± 1.3	1.77
TrisNHMAm	79.9 ± 4.8	72.3 ± 1.7	1.10
DMAm	72.0 ± 3.0	48.8 ± 0.9	1.48
TrisNHMAm + DMAm (A)	85.5 ± 1.6	63.4 ± 1.1	1.30
NHEAm + DMAm (B)	63.2 ± 0.6	50.6 ± 2.2	1.30

^aResults for comonomer combinations A (TrisNHMAm + DMAm) and B (NHEAm + DMAm) are also shown.

for myoglobin recognition, with an excellent 98.9% rebind and an imprinting factor of 1.90. The worst performing MIP was DMAm, with 72.0% rebind and an imprinting factor 1.48. With respect to overall MIP efficacy, the monomers could be ordered by performance NHMAm > AAm > NHEAm > TrisNHMAm > DMAm. The monomer NHEAm was judged to be better than TrisNHMAm on the basis of IF (1.77 versus 1.10, respectively), despite having a slightly lower % rebind (77.2% versus 79.9%, respectively).

The self-assembly method for MIP formation relies upon a degree of association between the monomer and the template, mainly based on hydrogen-bond interactions.⁹ Monomers containing –OH functional groups (NHMAm, NHEAm, and TrisNHMAm), therefore, have the potential to bind stronger than the monomers AAm and DMAm. In accordance with this, TrisNHMAm and NHMAm NIPs have the highest protein binding values. However, if one ranks the monomers by the MIP percentage protein rebind, then we have NHMAm > AAm > TrisNHMAm > NHEAm > DMAm. Thus, in terms of MIP, the AAm monomer without any –OH groups is the second best performing monomer, superior to both NHEAm and TrisNHMAm. In addition to this, TrisNHMAm with three –OH functional groups does not perform particularly well with a percentage rebind of 79.9%, only performing slightly better than DMAm (72.0%). TrisNHMAm also has an unsatisfactory IF value of 1.10, making it unsuitable for use as a myoglobin targeted MIP. This indicated that there are other factors, and not just the potential strength of protein–template interactions, to consider when choosing a suitable monomer for a target protein.

3.1.2. SiteMap Prediction of Myoglobin Binding Sites. Following preparation of the crystal structure of myoglobin for calculations, the potential binding sites were predicted using SiteMap, producing a map of 14 different binding sites. Visual inspection revealed that the 14 binding sites (1–14) were quite evenly distributed around the protein surface, as shown in Figure 2. This can be interpreted as favorable for MIP target protein binding, with monomers potentially more uniformly distributed around the template protein, compared to MIPs where the monomers are isolated to a specific area.³³

3.1.3. Docking Results for Monomer Binding. Glide-SP docking of all five monomers into each of the 14 binding sites was performed, with the best GlideScores of monomers at each site presented in Table S2. The only monomer, which successfully docked to all 14 sites, is the smaller AAm monomer. NHMAm docks into 13 binding sites, TrisNHMAm docks into

12 binding sites and the last two monomers (NHEAm and DMAm) dock into 11 binding sites. Hence, the best performing monomers experimentally as MIPs (NHMAm and AAm) docked into the most binding sites. TrisNHMAm, however, had a number of unfavorable GlideScores at different sites. The docking calculations revealed key hydrogen bonding interactions between the monomers and the protein molecule. This involved either monomer carbonyl, amine, or hydroxyl groups with similar functional groups on protein residues. Importantly, however, these hydrogen bond interactions were with both protein backbone and side chain atoms, the degree of which with protein backbones can be significant.³⁴ Prior to detailed analysis, more accurate binding predictions were performed using the docking poses and the theoretically more rigorous postdocking MM-GBSA method,⁵⁰ whose connection with statistical thermodynamics has been established.⁵¹

3.1.4. MM-GBSA Results for Monomer Binding. For each site, postdocking MM-GBSA binding free energies (eq 1) were calculated using the Glide-SP docking poses. The best values for each monomer and site are shown in Table 2. Breakdown of the contributions as per eq 1 are shown in Table S3. As explained in the methods, these values incorporate contributions due to protein-monomer interactions in the binding conformation (the MM term), solvent effects (the GB SA term), monomer reorganization and changes in monomer entropy upon binding, evaluated by the RRHO approximation.

Binding of the various monomers at the different sites is associated with a range of free-energy values ΔG_{PL} . Sites 2, 3, 5, 8, and 10 are associated with lower (more negative) values, ranging from ~ -4 to -12 kcal/mol (exception TrisNHMAm, *vide infra*), and generally appear to be key binding sites. Analysis of the ΔG_{PL} data from Table 2 revealed greater magnitude of values for monomers with more hydroxyl groups. TrisNHMAm (three hydroxyls) has the most negative ΔG_{PL} values and demonstrates an especially strong preference for protein binding site 10 ($\Delta G_{PL} = -24.9$ kcal/mol). NHEAm and NHMAm, each with one hydroxyl, consistently have better (more negative) ΔG_{PL} values compared to AAm (no hydroxyls); DMAm, also lacking hydroxyls, has slightly better ΔG_{PL} values than NHMAm and slightly worse than NHEAm.

Figure 3 displays the average number of bound monomers (prior to polymerization) as a function of the free monomer concentration $\alpha = [L]/C_0$, computed with eq 3. The largest value of the α -axis corresponds to the experimental total monomer concentration (0.76 M). Monomers with lower (more negative) binding free energies ΔG_{PL} are expected to bind with larger average numbers across the different binding sites. TrisNHMAm has the most monomers bound on the protein surface and AAm the least at all concentrations. The relative ranking depends somewhat on the monomer concentration, reflecting the fact that each monomer has a different distribution of binding free energies. For example, the monomer ranking at very small monomer concentrations (TrisNHMAm > NHEAm > NHMAm $\sim$ DMAm > AAm) is determined by the number of binding sites with $\Delta G_{PL} \leq -10$ kcal/mol, (six, two, one, one and zero, respectively, as shown in Table 2). DMAm becomes a better binder than NHMAm and NHEAm at higher concentrations, because it has a better distribution of binding free energies with less negative values ($-10 < \Delta G < 0$ kcal/mol).

Using eq 2 and the binding free energies reported in Table 2, the expected free monomer concentrations and the corresponding average number of bound monomers (prior to

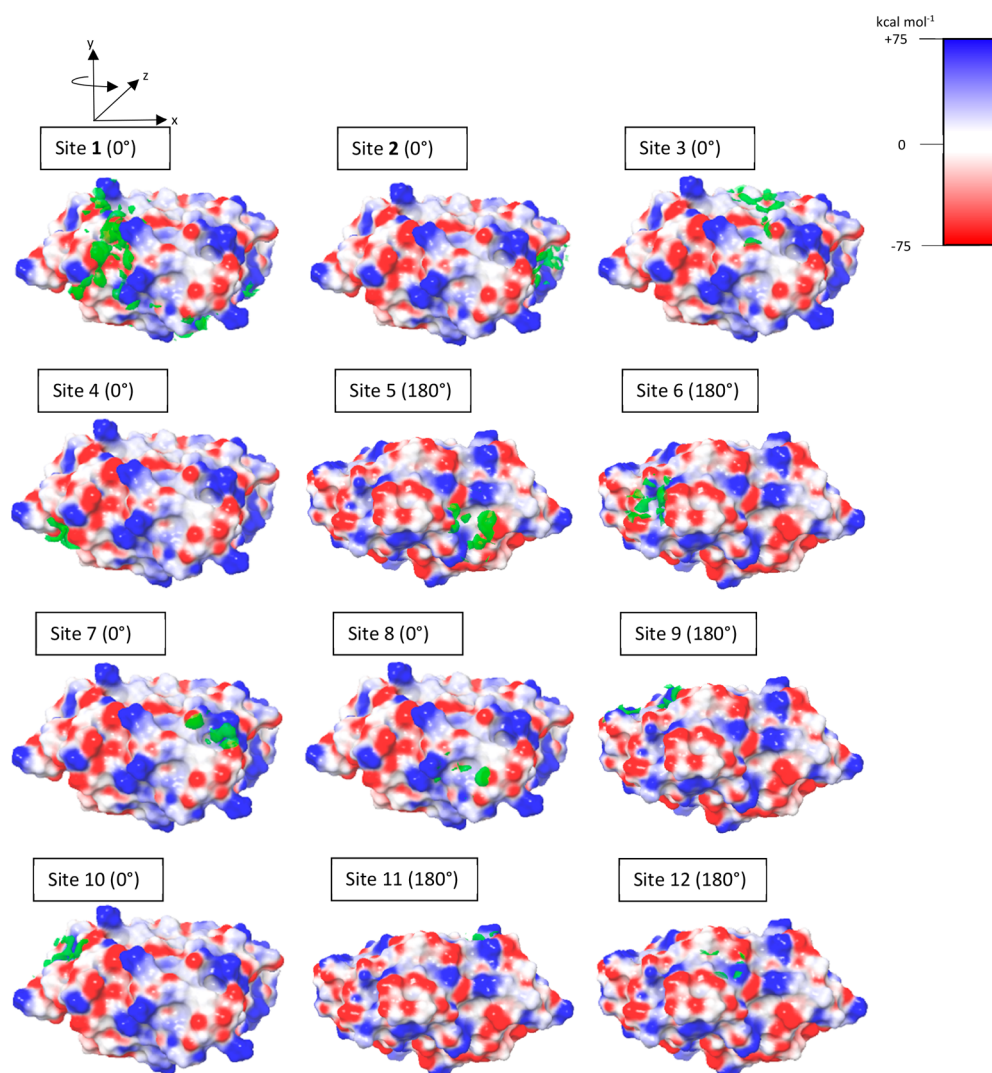


Figure 2. Potential monomer binding sites 1–12 (green) on myoglobin as predicted by SiteMap, together with the electrostatic potential (blue (positive) and red (negative)) mapped on the protein molecular surface calculated using Maestro. Sites 13 and 14 are not displayed as they were buried and not on the protein surface; it transpires they were predicted as not favorable for monomer binding (Table 2).

polymerization) were computed for each solution (Table 3). Due to the large monomer:protein ratio (1081:1), the free-monomer concentrations are similar to the total monomer concentration (0.76 M). Thus, the average number of bound monomers correspond to the values at the highest-concentration end of the curves in Figure 3. The relative ordering of our computed $\bar{M}_L(x)$ values, AAm < NHMAm < DMAm < NHEAm < TrisNHMAm, roughly reproduces the NIP efficiency of the various monomers (Table 1): NHEAm < AAm < DMAm < NHMAm < TrisNHMAm. This is reasonable, as the ability of the protein to bind prepolymerized individual monomers should be correlated with the nonimprinted cavity free polymer NIP.⁴⁶

Even though the free and total monomer concentrations are approximately equal, the computed estimates of Table 3 imply that the large majority of protein binding sites are occupied by monomers. For example, TrisNHMAm binds an average of 11.5 monomers in 12 sites with negative free energies; AAm seems to have the worst performance, with 9.17 monomers in 14 sites (average binding numbers depend strongly on the ΔG_{PL} values). The reported values are based on several approximations. Equations 2 and 3 assume that the various

sites are independent from each other. It is likely that the MM-GBSA approximation yields too negative binding free energies, as also observed elsewhere.^{52,53} Furthermore, below we show through CD spectroscopy that with the exception of AAm, binding of all other monomers modifies to a similar extent the protein secondary structure. In the MM-GBSA analysis, we neglect energy contributions due to protein reorganization; their inclusion increases the error in the binding free energies particularly when different binding sites are considered. It has been shown that relative binding free energies for congeneric series of ligands binding at proteins can be better reproduced when protein reorganization is neglected (presumably cancellation of errors).⁵⁴ Due to these approximations, it is likely that the obtained average numbers are only indicative but reflect the important relative trends among the different monomers.

3.1.5. Protein Surface Analysis. While monomers with more OH groups produce more favorable negative ΔG_{PL} values, this does not follow the experimental trend for monomer related MIP performance. It is clear that other factors such as a closer analysis of the binding interactions involved, as well as protein stability in the MIP prepolymerization mixture need consideration.

Table 2. ΔG_{PL} values (kcal mol^{-1}) of the Five Monomers (AAM, NHMAM, NHEAM, DMAm, TrisNHMAM) Calculated Using MM-GBSA at Each of the 14 Predicted Binding Sites of Myoglobin^a

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Monomer														
AAM	2.5	-9.0	-3.8	-2.2	-7.3	-2.8	1.6	-3.3	-1.2	-3.6	0.1	-1.7	7.3	9.2
NHMAM	-3.4	-10.8	-5.1	16.2	-9.8	-2.9	0.7	-5.3	-4.8	-9.9	-0.7	0.5	36.8	n/a
NHEAM	n/a	-11.3	-7.0	-5.9	-8.6	-3.2	-0.5	-6.6	-4.4	-12.9	6.9	-3.2	n/a	n/a
DMAm	n/a	-8.8	-9.1	-6.5	-7.3	-3.8	0.7	-5.4	-4.3	-10.7	-3.3	-2.3	n/a	n/a
TrisNHMAM	-10.3	-5.2	-10.7	-10.7	-12.1	-7.4	-0.5	-11.3	-1.3	-24.9	-5.2	-5.2	n/a	n/a
Comonomer														
MIP A	-10.3 (1.000)	-8.8 (0.998)	-10.7 (0.953)	-10.7 (0.999)	-12.1 (1.000)	-7.4 (0.998)	-0.5 (1.00)	-11.3 (1.00)	-4.3 (0.994)	-24.9 (1.00)	-5.2 (0.960)	-5.2 (0.992)	n/a (-)	n/a (-)
MIP B	n/a (-)	-11.3 (0.985)	-9.1 (0.971)	-6.5 (0.732)	-8.6 (0.898)	-3.8 (0.732)	-0.5 (1.00)	-6.6 (0.542)	-4.4 (0.500)	-12.9 (0.979)	-3.3 (0.858)	-3.2 (0.793)	n/a (-)	n/a (-)

^aFor comonomer mixtures MIP A (TrisNHMAM & DMAm) and MIP B (NHEAM & DMAm), the predicted ΔG_{PL} values given are for the monomer most probable to bind at the site, together with its probability value for binding (in parentheses) as calculated using eq 6. Values that are positive (unfavorable binding) are highlighted in bold. n/a indicates that there were no predicted binding poses.

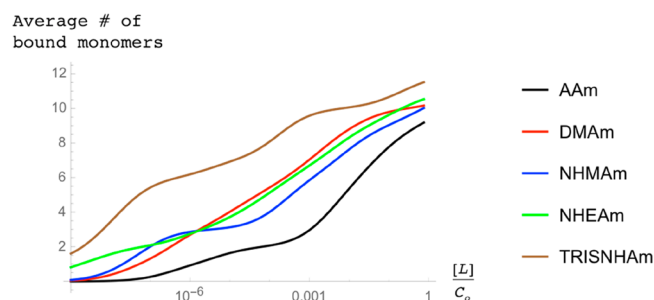


Figure 3. Average number of bound monomers as a function of the free monomer concentration $x = [L]/C_0$, computed with eq 3. The x -axis is shown in a logarithmic scale. The largest concentration corresponds to the total monomer concentration (0.76 M) at which point the average numbers are much closer.

Protein–monomer prepolymerization complexes can potentially force proteins into conformations or aggregates, which when imprinted produce MIPs that lack the desired target selectivity. The monomers interacting with the protein disrupt the secondary structure within the protein, particularly if these interactions are with protein backbone atoms.³⁴ On this basis, we have analyzed the surface of myoglobin using BioLuminate for the different amino acids involved in hydrogen bonding with the different monomers and whether these interactions are with backbones/side chains of amino acid residues in helices, β -strands and loops. The surface of the protein is mainly of a positive nature (positive surface area $4062 \text{ \AA}^2$; negative surface area $2116 \text{ \AA}^2$), but the overall formal charge on the protein was 0 (pI myoglobin = 6.8). In terms of potential hydrogen bonding, the donor surface area ($1434 \text{ \AA}^2$) exceeds the acceptor surface area ($1035 \text{ \AA}^2$), indicating that hydrogen bonding to acrylamide analogue carbonyl groups could be similarly favored. The residues on the protein surface that are positively and negatively charged, polar, hydrophobic and small nonpolar are highlighted in the FASTA primary structure sequence shown in Figure 4A, further highlighting the charged/polar nature of the surface. Figure 4B reveals that positively charged Lys residues (11) are the most favored for protein–monomer hydrogen bonding but involvement of the acidic Glu (9) and Asp (6) residues are common as well. The hydrogen bonding with these is mainly side chain, although some backbone interactions still occur. The rest of the residue types have ≤ 3 amino acids involved in hydrogen bonding interactions. Some nonpolar residues are also on the surface but by nature of their structure, only have backbone hydrogen bond interactions. The contributions of hydrogen bonding energies to ΔG_{PL} values are shown in Table S4 and range from -0.3 to -2.5 kcal/mol ; however, there is not much variance in their magnitudes between the different sites.

A more monomer-specific analysis of protein–monomer hydrogen bonding is shown in Table 4. The TrisNHMAM monomer has the highest number (42) of total hydrogen bond interactions with the protein and, significantly, also the highest number (8) of helical backbone interactions. It is these interactions, in particular, that can affect protein secondary structure and have a negative impact on MIP selectivity.³⁴ So, while monomers with hydroxyl groups such as TrisNHMAM can form stronger interactions with protein molecules, these interactions do not necessarily lead to a better MIP due to potential changes in the secondary structure caused by the additional hydrogen bonds. Parts A and B of Figure 5 show the binding of TrisNHMAM and AAM at binding site 5,

Table 3. Free Monomer Concentration and Average Number of Monomers Bound to the Protein for Each Solution

monomer/comonomer solution	concentration of free monomer ^a x (mol/L)	average number of bound monomers ^b $\bar{M}_1(x)$
AAm	0.7536	9.17
NHMAm	0.7530	10.0
NHEAm	0.7526	10.5
DMAm	0.7529	10.2
TrisNHMAm	0.7519	11.5
DMAm/TrisNHMAm	0.3785/0.3734 ^c	2.14/9.34 ^c
DMAm/NHEAm	0.3766/0.3761 ^c	4.85/5.62 ^c

^aComputed from eq 2 (single monomer solutions) and eq 4 (comonomer solutions). ^bComputed from eq 3 (single monomer solutions) and eqs 5a, 5b (comonomer solutions), using the free monomer concentration of column 2. ^cThe numbers correspond to the first and second listed monomer, respectively.

respectively. AAm forms only hydrogen bonds with side chains, whereas TrisNHMAm is involved in a network of interactions, some of which are backbone. DMAm is predicted to have relatively few hydrogen bond interactions (14) but a high proportion are hydrogen bonds with the helical backbones (4). As DMAm does not have any hydrogen bond donors, these are all with DMAm as hydrogen bond acceptor and the hydrogen bonding energies of DMAm are generally of smaller magnitude (Table S4). Hence, unlike all other monomers, DMAm is

Table 4. Summary of the Predicted Hydrogen Bonding Interactions for Each of the Five Monomers (Figure 1) with Myoglobin As Calculated Using MM-GBSA^a

Monomer	helical interactions		loop interactions		total
	backbone	side chain	backbone	side chain	
AAm	5	9	4	4	22
NHMAm	3	15	5	4	27
NHEAm	1	12	3	6	22
DMAm	4	5	2	3	14
TrisNHMAm	8	23	5	6	42
Comonomer ^b					
MIP A	5	18	3	8	34
MIP B	6	9	4	4	23

^aFor comonomer MIPs A (TrisNHMAm and DMAm) and B (NHEAm and DMAm), these interactions were estimated on the basis of the dominant binding monomer at a particular site (Table 2). Protein contained no β -sheet secondary structure hydrogen bond interactions. ^bInteractions based on dominant binding monomer for each site (Table 2). In case of equal probabilities for monomers, the monomer with most interactions was considered.

unable to exploit hydrogen bond interactions with the many Glu/Asp side chains on the surface of the protein (Figure 4), also in line with its poor overall MIP performance (Table 1). NHMAm and NHEAm have a low proportion of helical backbone interactions (3 and 1, respectively) in comparison to their relatively large number of interactions (27 and 22, respectively). It is notable also that the best performing monomer as

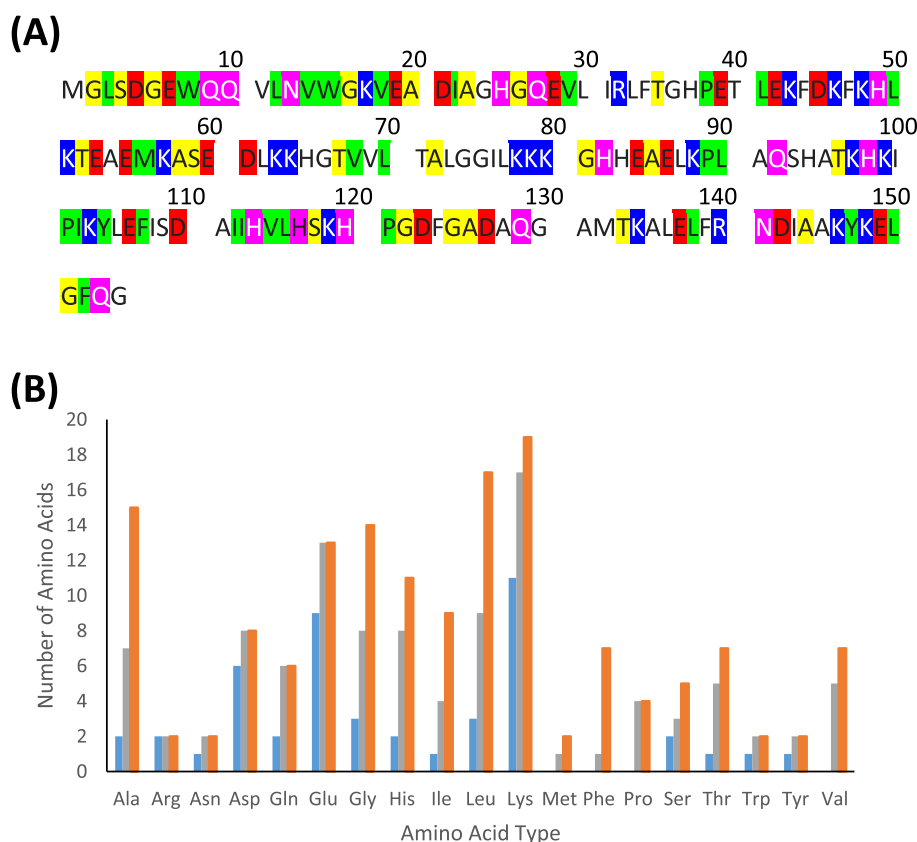


Figure 4. Myoglobin protein residue and surface analysis using Bioluminate. (A) FASTA primary structure sequence with the positive, negative, polar, hydrophobic, and small nonpolar surface residues shown in blue, red, magenta, green, and yellow, respectively. (B) Total number of the different types of amino acids in the protein (orange) compared to the number found on the surface of the protein (gray) and the number of these involved in hydrogen bond interactions with any monomer (blue).

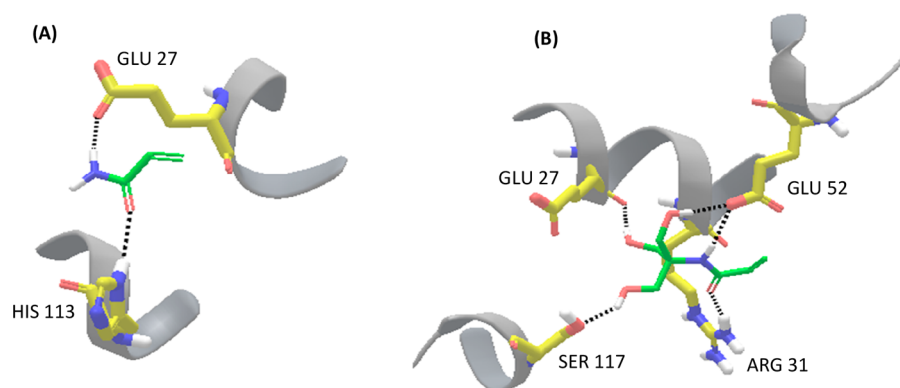


Figure 5. Hydrogen bond interactions between myoglobin residues at binding site 5 and the monomers (A) AAm and (B) TrisNHMAm. Whereas AAm only has side chain interactions, TrisNHMAm has a network of interactions with both backbone and side chain.

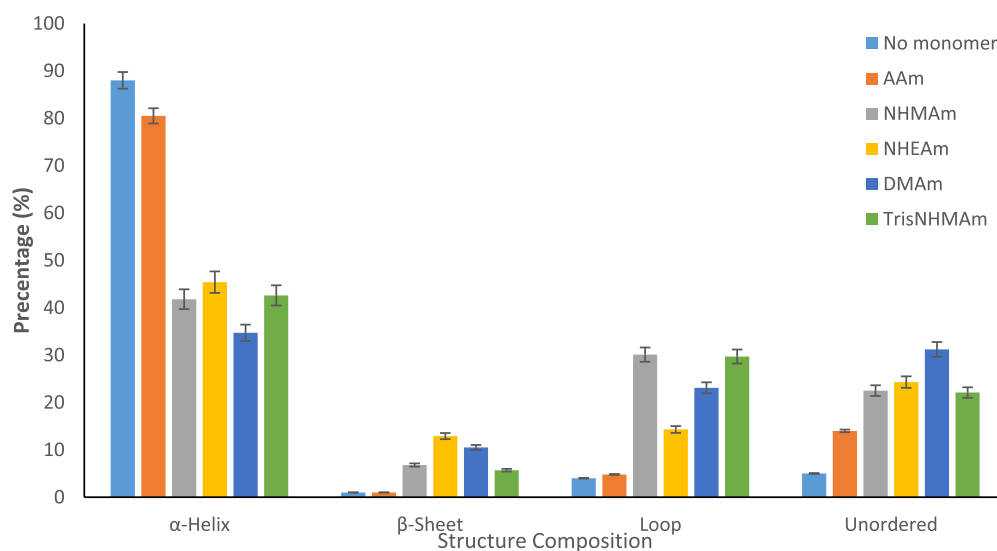


Figure 6. Secondary structure composition of myoglobin after being mixed with each of the five monomers (AAm, NHMAm, NHEAm, DMAm, TrisNHMAm) at a protein:monomer ratio of 1:1081, the same as the polymerization solution used in hydrogel MIP formation.

MIP (NHMAm) has the most hydrogen bond contacts overall, TrisNHMAm excepted.

3.1.6. CD Spectroscopy Result. The computational analysis revealed that the monomers potentially interact with protein helical backbone to varying degrees. As mentioned, this can cause changes in myoglobin secondary structure, leading to poor selectivity for target rebinding to the MIP. Accordingly, CD spectroscopy was used to explore how the different monomers can affect myoglobin secondary structure before polymerization, with the resulting percentage structural composition shown in Figure 6 and the corresponding spectra included in Figure S1.

The native structure of myoglobin was determined to be $88.0 \pm 2\%$ α -helix, $1.0 \pm 2\%$ β -sheet, $4.0 \pm 1\%$ turn, and $5.0 \pm 1\%$ unordered/random. Each monomer was added to a solution of the protein at a ratio of 1:1081, which is consistent with the experimental protein:monomer ratios in MIP prepolymerization mixtures. The monomer AAm produced a structural composition, which is similar to that of native myoglobin, with $80.5 \pm 2\%$ α -helix, $1.0 \pm 2\%$ β -sheet, $4.8 \pm 1\%$ turn, and $14.0 \pm 1\%$ unordered/random. While there is a slight decrease in α -helix composition, this result indicated a significant basis for AAm performing well in the target rebinding studies (Table 1). AAm did have some predicted helical backbone interactions

(Table 4) but it is likely that its smaller size and lack of competing hydrogen bond contacts with protein compared to, for example, TrisNHMAm (Figure 5B) may be a crucial factor. In fact, all other monomers lead to significant secondary structure changes. The monomer DMAm produced a structural composition of only $34.7 \pm 2\%$ α -helix, $10.5 \pm 2\%$ β -sheet, $23.1 \pm 1\%$ turn, and $31.2 \pm 1\%$ unordered/random. Hence, there is considerable change of the myoglobin native secondary structure features, suggestive of selectivity issues and another contributory factor for DMAm producing the worst performing monomer-based MIP (Table 1). DMAm is predicted to form relatively fewer hydrogen bonds and a large proportion were helical backbone. There is additionally potential steric effects due to the dimethyl group. The monomer TrisNHMAm also produced structural compositions with considerable changes in secondary structure, with $42.6 \pm 2\%$ α -helix, $5.7 \pm 1\%$ β -sheet, $29.7 \pm 1\%$ turn, and $22.1 \pm 1\%$ unordered/random observed, which could be expected on the basis of extensive hydrogen bonding interactions (three monomer hydroxyls; eight predicted helical backbone interactions). Somewhat surprisingly, however, the structurally similar NHMAm (three helical backbone interactions) and NHEAm (one helical backbone interaction) each with one hydroxyl, also cause secondary structure changes with values similar to

those of TrisNHMAm. These monomers did not have many helical backbone interactions, yet secondary structure was affected. It does appear, therefore, that monomer size is a factor. Despite this, NHMAm and NHEAm performed well in the MIP target rebinding experiments, with the NHMAm based MIP in fact being the top-performer. NHMAm is smaller and less flexible than the structure extended by the $-\text{CH}_2-$ analogue, NHEAm. In comparison, flexibility of cross-linker in MIP design is reported as favored.⁵⁵ In conclusion, the CD data have indicated that secondary structure alteration (or lack of it) is likely a key factor in MIP performance (AAm, TrisNHMAm, DMAm); however, NHMAm and NHEAm despite having a clear effect on secondary structure features still perform well in the MIP experiments.

Overall, analysis of the single monomer computational and experimental data unquestionably indicates a complex interplay of potential factors (including site binding free energies, nature of protein–monomer interactions, and secondary structure changes) for a successful monomer used for producing protein-targeted MIPs. This led us to consider combining features of different monomers in carefully selected comonomer experiments predicting outcomes in advance to validate our hypotheses.

3.2. Comonomer MIPs Studies. **3.2.1. Rational Design of Comonomer MIPs.** On the basis of the single monomer studies, comonomer MIPs were rationally designed, synthesized, and further experimental rebinding studies performed. The criteria we set for good performance were that monomers bind strongly across the different predicted binding sites. Additionally, predicted binding of the comonomer mixtures should have limited helical backbone interactions and have minimal competition with each other for each binding site. Two comonomer MIPs were selected for synthesis. The first MIP (comonomer mixture A) was designed to produce, in theory, a comonomer MIP with better performance than either of the MIPs for the individual monomers. The second MIP (comonomer mixture B) was predicted to perform worse than the individual comonomers MIPs. Equation 6 was employed to calculate the site-specific relative binding probabilities of monomers for the two comonomer solutions (Table 2). The average number of each monomer $\bar{M}_L(x)$ bound for the MIPs are shown in Table 3.

Comonomer mixture A contained a 50:50 mixture of the monomers TrisNHMAm and DMAm. This combination represented a good test of our hypotheses for improvement in MIP performance. The computed binding free energies suggested that DMAm binds strongly into the binding sites where TrisNHMAm does not (sites 2 and 9). This suggests that for MIP A, there will be potentially strong uniform binding to 12 of the 14 predicted binding sites. The TrisNHMAm:DMAm combination also takes into account the number of helical backbone interactions. Theoretically, a monomer is placed into a binding site, resulting in less helical hydrogen bonding, and therefore potential for less structural changes in the protein compared to TrisNHMAm alone (Table 4). The average number of bound monomer $\bar{M}_L(x)$ is expected to be mainly TrisNHMAm (9.34) compared to DMAm (2.14) due to the much more negative TrisNHMAm ΔG_{PL} values (Table 3).

Comonomer mixture B consisted of the monomers NHEAm and DMAm. These monomers were chosen based on predicted binding competition at sites 3, 4, 5, 6, 8, 9, 10, 11, and 12. The average number of bound monomers $\bar{M}_L(x)$ was expected to be similar for each monomer, NHEAm with 5.62 compared to DMAm (4.85) (Table 3). This competition of monomers for

binding sites reduces the selectivity of a MIP.³⁴ Also, for this combination, the sites where there is the strongest binding has considerable helical backbone binding (Table 4), and hence increasing the potential for secondary structure changes. Overall, therefore, we expected a less effective comonomer MIP B compared to those prepared using the individual NHEAm and DMAm monomers.

3.2.2. Experimental Rebinding Comonomer Results. The percentage rebind and imprinting factors (IF) results for the comonomer MIPs A (TrisNHMAm and DMAm) and B (NHEAm and DMAm) and their corresponding NIPs are shown in Table 1.

MIP A with a 1:1 ratio of TrisNHMAm and DMAm produced a MIP rebind of 85.5%, which was much better percent rebind than when either of the individual monomers was used (79.9% and 72.0% for TrisNHMAm and DMAm, respectively). The percentage rebind for the NIP is 63.4%, leading to an IF value (1.3) greater than the established 1.2 threshold. In agreement with our predictions, therefore, this rationally designed combination of monomers to provide a more uniform coverage of the protein binding sites resulted in improved performance of the MIP. Meanwhile, MIP B produced a MIP rebind of 63.2%, considerably worse than each of the individual monomers (77.2% and 72.0% for NHEAm and DMAm, respectively). Again, in agreement with our predictions, this combination with predicted competition for binding sites led to decreased performance of the MIP compared to individual monomers. Exploiting computational and rational design such as employed here, therefore, we have demonstrated the potential for prediction of the performance of a MIP in advance is possible.

4. CONCLUSIONS

We have exploited computational methods to gain a further understanding of the performance of hydrogel-based protein imprinted MIPs, based on predictions of protein–monomer interactions at the atomic level. Identification of 14 potential binding sites on the chosen target protein myoglobin was followed by molecular docking of five acrylamide-based test monomers (AAm, NHMAm, NHEAm, TrisNHMAm, DMAm). MM-GBSA postdocking calculations were performed to calculate the binding free energies and used to assess monomer performance. The results revealed that monomers with hydroxyl functional groups (NHMAm, NHEAm, TrisNHMAm) have generally stronger binding free energy (ΔG_{PL}) values than those without the hydroxyl groups (AAm). While this showed some correlation with experimental data and with the NIP performance of the various monomers, it did not reflect, for example, the better MIP performance of the AAm monomer, with few protein–ligand hydrogen bonds (smaller ΔG_{bind} magnitudes) and the poorer performance of TrisNHMAm, with many hydrogen bonds (larger ΔG_{PL} absolute values). Analysis of protein–monomer interactions helped in part to explain these anomalies. In comparison to AAm, TrisNHMAm had a large amount of (interhelical) protein backbone interactions, which could result in the secondary structure composition changes observed by CD spectroscopy, reducing the selectivity of the MIP. The monomers NHEAm and NHMAm despite affecting myoglobin secondary structure did produce MIPs with good selectivity, with the better MIP performance of NHMAm over NHEAm potentially related in part to less conformational flexibility. In fact, the smallest monomers AAm and NHMAm produced the best performing

MIPs indicating that this simple parameter is an important factor. Analyzing predictions for binding of monomers at the individual sites (in terms of site specific ΔG_{PL} values and protein-monomer interactions), two comonomer MIPs, MIP A, and MIP B were rationally designed. For MIP A, monomers predicted to increase uniform binding around the surface of the protein were selected, while minimizing helical backbone interactions and competition for binding sites. This resulted in a comonomer MIP that performed better than the individual MIPs. For MIP B, rationally designed to perform less well than the individual monomers, two monomers that had competition for binding sites and that interacted strongly with the protein backbone and helical structure were selected, all of which should reduce MIP performance. Again, experimental validation was in line with the predictions. Overall, therefore, a better understanding of key atomistic interactions within a MIP–protein complex has been established in this study and represents a further step toward the rational design of MIPs with predictable characteristics in the future.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.9b03091.

Detailed derivation of the equations used for the thermodynamic analysis; tables of Glide protein grid coordinate centers, GlideScore values, breakdown of ΔG_{PL} into its contributions, ΔE_{Hbond} contributions; figures of CD spectra, (PDF)

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Notes

The authors declare no competing financial interest.

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